Hall Ticket Number:

MR-22

Code: B325CM2S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July 2025

Software Engineering

(Computer Science and Engineering (AI and ML))

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1= 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write about characteristics of software?	1	2	1	1
1. b)	List the Different types of software myths?	1	1	1	1
1. c)	Simplify interface and write about interface specification?	1	4	1	1
1. d)	Discuss the importance of feasibility study?	1	2	1	3
1. e)	List out the design concepts?	1	4	2	1
1. f)	Discuss about class diagrams?	1	6	2	1
1. g)	Write about unit testing?	1	2	3	3
1. h)	Simplify the factors that affect quality?	1	4	2	1
1. i)	Write about the two characteristics of software 7risks?	1	2	3	2
1. j)	Discuss about software reviews?	1	2	2	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the changing nature of software?	5	2	1	1
2. b)	Discuss about CMMI in detail?	5	3	1	1
	OR				
2. c)	Explain about Software process framework?	5	2	1	1
2. d)	Compare waterfall model with spiral model?	5	4	1	1
3. a)	Explain both functional and non-functional requirements?	5	2	2	1
3. b)	Discuss the process of preparing software requirements document?	5	3	2	2
	OR				
3. c)	Discuss about requirements management?	5	2	2	2
3. d)	Write about requirements elicitation and analysis in detail?	5	1	2	3
4. a)	List and explain different architectural styles?	5	2	2	2
4. b)	Explain the design concepts?	5	3	2	3
	OR				
4. c)	What is UML write about the applications of UML?	5	3	2	3
4. d)	Develop and explain examples for Usecase and sequence diagrams?	5	4	1	3
5. a)	Discuss different types of Black box testing techniques?	5	2	3	5
5. b)	Explain in detail about integration testing?	5	3	3	5
	OR				
5. c)	Explain about function oriented metrics?	5	2	2	3
5. d)	Explain about the art of debugging?	5	2	3	4
6. a)	Explain the process of preparing the risk table?	5	2	3	2
6. b)	Write About RMMM with a suitable example?	5	3	3	1
	OR				
6. c)	Explain about Formal Technical Reviews?	5	2	3	1
6. d)	Discuss about SQA in detail?	5	2	2	5

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome